Interrogator System

Progressive challenges that escalate based on response — let humans prove themselves, trip up bots. The goal isn't to block everyone suspicious. It's to create asymmetric friction.

3	5-Level	0	Async
CHALLENGE LEVELS	ESCALATION PATH	BACKEND REQUIRED	FRICTION MODEL

CHALLENGE ESCALATION FLOW

Incoming suspicious request · Risk 40–70 · Not block, not allow

L1 EASY

Cookie Challenge

Set-Cookie header · Check if client returns it

CATCHES: curl, wget, simple scripts — anything that ignores cookies

✓ PASS → Allow
✗ FAIL → Level 2
Risk −10 on pass

↓

L2 MEDIUM

JS Proof-of-Work

SHA-256 computational challenge · Client computes hash

CATCHES: headless browsers, basic bots — can't execute JavaScript or fail PoW

✓ PASS → Allow
✗ FAIL → Level 3
Risk −15 on pass

↓

L3 HARD

CAPTCHA

Image recognition · Human interaction required

CATCHES: sophisticated automation — the last gate before enforcement

✓ PASS → Allow
✗ FAIL → Block + Tarpit
Risk −25 on pass (verified human)

↓

BLK

BLOCK + TARPIT

All challenges failed. Connection held open with artificial delays (1–30s). Each failed challenge adds +20 risk. Retry window shrinks progressively. Attack becomes economically unviable.

TARPIT SYSTEM

After all challenges fail: asymmetric resource drain

Resource Drain

Connection held open 1–30 seconds. Attackers must wait, consuming their resources — not ours.

Progressive Delay

Each subsequent attempt from the same actor gets a longer delay. Retry windows shrink exponentially.

Economic Impact

Attack becomes economically unviable. Their time and infrastructure are wasted. Attackers move on to easier targets.

TWO PATHS — ASYMMETRIC FRICTION

✓ HUMAN / LEGITIMATE USER

Browser accepts cookie → Level 1 passes → Access granted

Most users never see Level 2 or 3. Friction is minimal — cookie set/return is invisible to the user. Risk reduction: −10 to −25 points based on level passed.

Friction: Minimal

✗ BOT / ATTACKER

Script ignores cookies → JS fails → CAPTCHA fails → Blocked

Each failed challenge adds risk, reduces retry window. Tarpits make retry economically painful. Risk increase: +20 per failed challenge, eventual permanent block.

Friction: Maximum

RISK SCORE IMPACT

−10	−15	−25	+20
Cookie pass (L1)	JS PoW pass (L2)	CAPTCHA pass (L3)	Per failed challenge

Humans pass quickly. Bots get stuck, slowed, and eventually blocked.

The goal is asymmetric friction — make attacking economically unviable without impacting legitimate users.